Chapter 7. Use Case Descriptions

UC01 — Register

Use Case ID	UC01
Use Case Name	Register
Module	Customer
Objective	Allow a new customer to create an account so they can request deliveries and track shipments.
Primary Actor	Customer
Secondary Actor	System
Preconditions	The customer is not yet registered and has a valid phone number/email.
Trigger	Customer selects "Register" on the Customer App.
Postconditions	A customer account is created and the customer is logged in.

Main Flow

1. Customer opens the Register screen.
2. Customer enters name, phone number, email and password.
3. System validates the input and checks for duplicate phone/email.
4. System sends an OTP to the phone/email.
5. Customer enters the OTP.
6. System verifies the OTP and creates the account.
7. System shows a success message and navigates to the Home screen.

Alternative Flow

- A1. Customer registers using a social/third-party login, skipping password entry.

Exception Flow

- E1. Phone/email already registered → show error and offer to log in instead.
- E2. OTP incorrect or expired → allow resend up to a configured limit.

Business Rules

- BR14 Notification
- BR16 Data Privacy

Related Screens

CUS-08	Register
CUS-09	OTP Verification

Main Data Fields

customer_id, full_name, phone_number, email, password_hash, otp_code, account_status.

Notes

Registration is a catalog use case; keep the flow minimal. OTP channel (SMS/email) is configurable.

UC02 — Login

Use Case ID	UC02
Use Case Name	Login
Module	Customer / Counter Staff / Dispatcher / Driver
Objective	Authenticate a user and grant access to the features allowed for their role.
Primary Actor	Customer / Counter Staff / Dispatcher / Driver
Secondary Actor	System
Preconditions	The user has an active account.
Trigger	User selects "Login".
Postconditions	The user is authenticated with a valid session.

Main Flow

8. User opens the Login screen.
9. User enters phone/email and password.
10. System authenticates the credentials.
11. System loads the role and permitted features.
12. System navigates the user to the role-specific home screen.

Alternative Flow

- A1. User logs in with biometric/social login if previously enabled.

Exception Flow

- E1. Wrong credentials → show error, allow retry, lock after configured attempts.
- E2. Account disabled → show message to contact support.

Business Rules

- BR15 Access Control
- BR17 Audit Logging

Related Screens

CUS-10	Login
SYS-01	Forgot Password

Main Data Fields

user_id, role, credential, session_token, last_login_at.

Notes

Single login use case serves all human actors; the destination screen depends on role.

UC03 — Create Delivery Request

Use Case ID	UC03
Use Case Name	Create Delivery Request
Module	Customer
Objective	Let a customer create a delivery request with one or more pickup points and a single destination, choosing a service level.
Primary Actor	Customer
Secondary Actor	System
Preconditions	Customer is logged in.
Trigger	Customer selects "Create Delivery".
Postconditions	A shipment is created with a tracking number and forwarded to dispatch.

Main Flow

13. Customer opens the Create Delivery screen.
14. Customer enters one or more pickup points (multi-pickup).
15. Customer enters a single destination point.
16. Customer selects a service level: Standard or Priority.
17. Customer enters parcel information and marks COD if applicable.
18. System validates the shipment (UC via Validate Shipment) and generates a tracking number (UC20).
19. If prepayment is required, the customer pays online (Payment); if COD, this step is skipped.
20. System confirms the request and forwards it to dispatch (UC07).

Alternative Flow

- A1. Walk-in: Counter Staff creates the request on behalf of a customer (UC06).
- A2. Priority order: system shows the surcharge and priority handling note.

Exception Flow

- E1. Invalid input (empty/badly formatted address) → show error and request correction.
- E2. Online payment fails → allow retry or switch to COD where permitted.

Business Rules

- BR01 Standard Delivery
- BR02 Priority Delivery
- BR03 Multiple Pickup Points
- BR09 Tracking Number

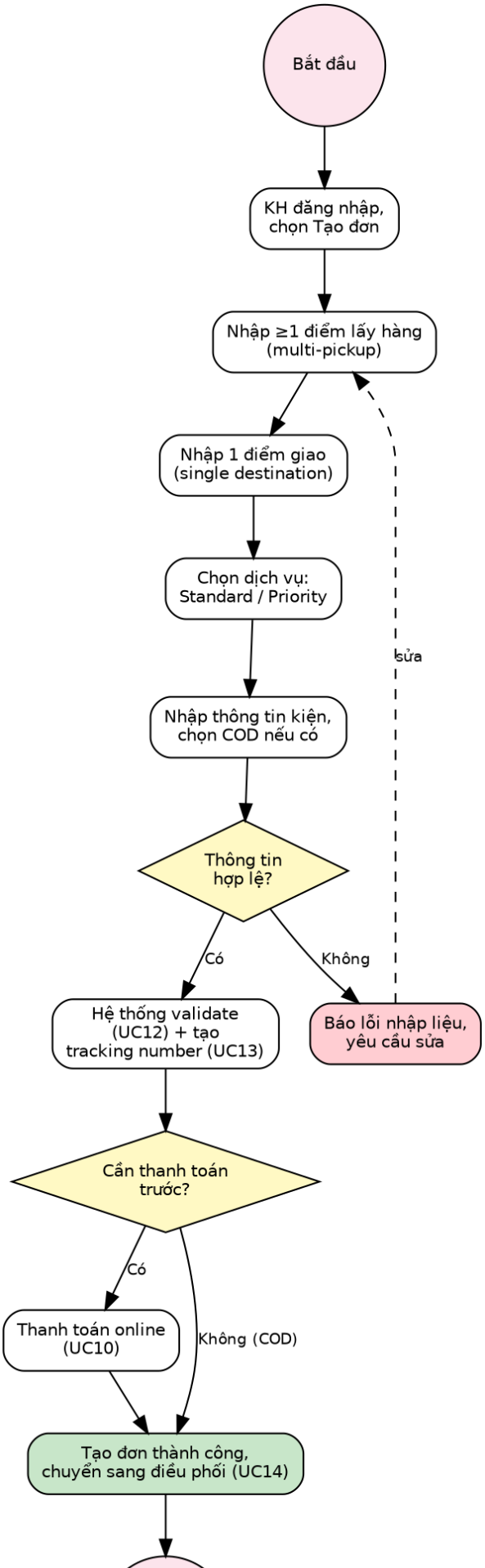
Related Screens

CUS-01	Home
CUS-02	Create Delivery
CUS-03	Pickup Information
CUS-04	Receiver Information
CUS-05	Confirmation

Main Data Fields

customer_id, pickup_points[] (>=1), dropoff_point, service_type (STANDARD/PRIORITY), parcel_info, cod_flag, cod_amount, tracking_number, payment_status.

Activity Diagram


Bắt đầu
KH đăng nhập, chọn Tạo đơn
Nhập ≥1 điểm lấy hàng (multi-pickup)
Nhập 1 điểm giao (single destination)
Chọn dịch vụ: Standard / Priority
Nhập thông tin kiện, chọn COD nếu có
Thông tin hợp lệ?
Có
Không
sửa
Hệ thống validate (UC12) + tạo tracking number (UC13)
Báo lỗi nhập liệu, yêu cầu sửa
Cần thanh toán trước?
Có
Không (COD)
Thanh toán online (UC10)
Tạo đơn thành công, chuyển sang điều phối (UC14)

Notes

A request allows multiple pickup points but exactly one destination. Priority is prioritised during dispatch and hub handling but does not guarantee a fixed delivery time.

UC04 — Track Shipment

Use Case ID	UC04
Use Case Name	Track Shipment
Module	Customer
Objective	Allow a customer to see the current status and location of their shipment in real time.
Primary Actor	Customer
Secondary Actor	System
Preconditions	The customer has at least one active shipment.
Trigger	Customer selects a shipment to track.
Postconditions	The customer sees the up-to-date shipment status.

Main Flow

21. Customer opens the Track Shipment screen.
22. Customer selects or enters a tracking number.
23. System retrieves the shipment status and timeline.
24. System displays the current status, latest location and estimated progress.

Alternative Flow

- A1. Customer receives a push notification and taps it to open the tracking screen directly.

Exception Flow

- E1. Tracking number not found → show a not-found message.
- E2. Location temporarily unavailable → show last known status only.

Business Rules

- BR10 Shipment Status
- BR14 Notification

Related Screens

CUS-06	Track Shipment

Main Data Fields

tracking_number, shipment_status, status_timeline[], current_location, estimated_progress.

Notes

Status values follow the shipment lifecycle defined in BR10.

UC05 — View Shipment History

Use Case ID	UC05
Use Case Name	View Shipment History
Module	Customer
Objective	Allow a customer to review their past and completed shipments.
Primary Actor	Customer
Secondary Actor	System
Preconditions	Customer is logged in.
Trigger	Customer opens the History screen.
Postconditions	The customer can review historical shipments.

Main Flow

25. Customer opens the Shipment History screen.
26. System retrieves the customer's completed and cancelled shipments.
27. System displays the list ordered by date.
28. Customer selects a shipment to view its detail.

Alternative Flow

- A1. Customer filters history by date range or status.

Exception Flow

- E1. No history yet → show an empty-state message.

Business Rules

- BR10 Shipment Status

Related Screens

CUS-07	Shipment History

Main Data Fields

customer_id, shipment_list[], shipment_status, created_at, completed_at.

Notes

Read-only screen; reuses the shipment detail view from tracking.

UC06 — Create Walk-in Shipment

Use Case ID	UC06
Use Case Name	Create Walk-in Shipment
Module	Counter Staff
Objective	Let counter staff create a delivery request on behalf of a walk-in customer who does not use the app.
Primary Actor	Counter Staff
Secondary Actor	System
Preconditions	Counter Staff is logged in to the Counter Portal.
Trigger	A walk-in customer requests a delivery at the counter.
Postconditions	A shipment is created for the walk-in customer and forwarded to dispatch.

Main Flow

29. Counter Staff searches for an existing customer or creates a quick profile.
30. Counter Staff enters pickup point(s) and the destination.
31. Counter Staff selects the service level and parcel information.
32. Counter Staff marks COD/prepaid and collects payment if prepaid.
33. System validates the shipment and generates a tracking number (UC20).
34. Counter Staff prints/【hands over】the tracking number to the customer.
35. System forwards the shipment to dispatch (UC07).

Alternative Flow

- A1. Returning walk-in customer is found by phone number and reused.

Exception Flow

- E1. Payment not completed for a prepaid order → shipment is held as draft.

Business Rules

- BR01 Standard Delivery
- BR02 Priority Delivery
- BR03 Multiple Pickup Points
- BR09 Tracking Number

Related Screens

CS-01	Walk-in Shipment
CS-02	Customer Lookup

Main Data Fields

counter_staff_id, customer_ref, pickup_points[], dropoff_point, service_type, parcel_info, payment_method, tracking_number.

Notes

This use case <<extend>>s UC03; the core creation logic is shared.

UC07 — Determine Delivery Strategy

Use Case ID	UC07
Use Case Name	Determine Delivery Strategy
Module	Dispatcher / System
Objective	Automatically decide whether a shipment is delivered directly (Direct) or routed through a hub (Hub).
Primary Actor	System (Dispatch Engine)
Secondary Actor	Dispatcher
Preconditions	The shipment is created and validated; Maps data is available.
Trigger	A shipment is created and validated (after UC03/UC06).
Postconditions	delivery_strategy is set to DIRECT or HUB and the shipment moves to the corresponding dispatch path.

Main Flow

36. Receive the validated shipment (pickup, destination, service level).
37. Compute pickup-to-destination distance via Maps API.
38. Determine the service area of the pickup and destination points.
39. Check whether both points fall within one hub/one driver coverage.
40. Check whether the distance is within the configured threshold.
41. Check whether a driver has enough remaining shift time to run the full leg.
42. If all conditions are met set strategy = DIRECT; otherwise set strategy = HUB (call UC18 to recommend a hub).
43. Store the delivery_strategy on the shipment and forward to the matching step.

Alternative Flow

- A1. Priority order: the system may prefer Direct when the distance is near the threshold, to reduce time.

Exception Flow

- E1. Maps distance unavailable → use straight-line distance temporarily and flag for review.
- E2. Service area cannot be determined → escalate to Dispatcher for manual handling.

Business Rules

- BR06 Hub Routing
- BR01 Standard Delivery
- BR02 Priority Delivery

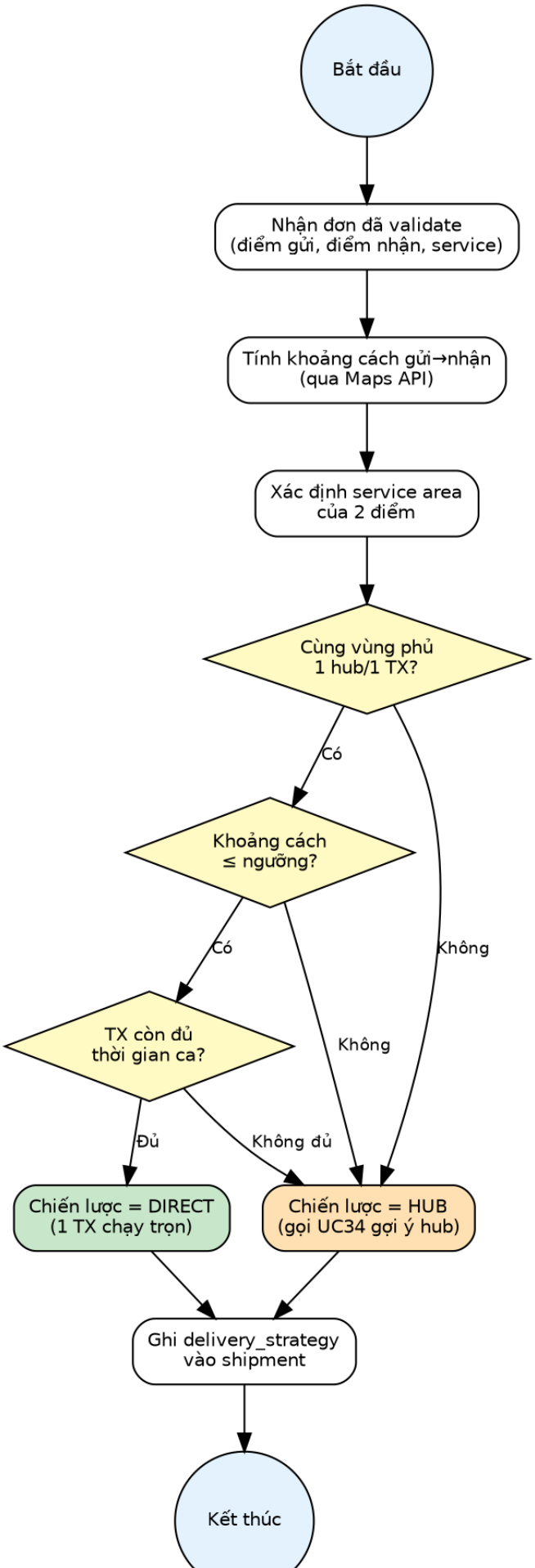
Related Screens

DSP-01	Dispatch Board
DSP-05	Routing Configuration

Main Data Fields

shipment_id, pickup_location, dropoff_location, service_type, distance_km, pickup_service_area, dropoff_service_area, delivery_strategy.

Activity Diagram


Bắt đầu
Nhận đơn đã validate
(điểm gửi, điểm nhận, service)
Tính khoảng cách gửi→nhận
(qua Maps API)
Xác định service area
của 2 điểm
Cùng vùng phủ
1 hub/1 TX?
Có
Khoảng cách
≤ ngưỡng?
Có
Không
TX còn đủ
thời gian ca?
Không
Đủ
Không đủ
Chiến lược = DIRECT
(1 TX chạy trọn)
Chiến lược = HUB
(gọi UC34 gợi ý hub)
Ghi delivery_strategy
vào shipment
Kết thúc

Notes

Runs mostly in the background; its result is visible on the Dispatcher board. Thresholds are configurable, not hard-coded.

UC08 — Assign Driver

Use Case ID	UC08
Use Case Name	Assign Driver
Module	Dispatcher / System
Objective	Assign a shipment or a leg to the most suitable driver using a multi-factor matching score, with batching of same-route orders.
Primary Actor	System (Dispatch Engine)
Secondary Actor	Dispatcher
Preconditions	At least one driver is eKYC-verified, checked-in and has enough shift time.
Trigger	A shipment or leg needs a driver (Direct, first-mile, or last-mile at a hub).
Postconditions	driver_id is assigned to the leg and the driver is notified.

Main Flow

44. Receive the assignment request for a shipment/leg.
45. Filter drivers: eKYC-verified (UC10), checked-in (UC11), enough shift time.
46. Compute the matching score: distance, traffic, current load, rating.
47. Check whether same-route orders can be batched.
48. Select the highest-scoring driver.
49. Send the offer to the driver (call UC19 Notify).
50. If the driver accepts within the time window, record driver_id for the leg.

Alternative Flow

- A1. Batching: if two same-direction orders qualify, assign both to one driver as a single route.
- A2. Dispatcher override: the dispatcher may assign a driver manually.

Exception Flow

- E1. No eligible driver → move to UC09 Reassign or escalate.
- E2. Driver rejects or times out → move to UC09 Reassign Driver.

Business Rules

- BR04 Driver Assignment
- BR07 Driver eKYC
- BR11 Shift Availability

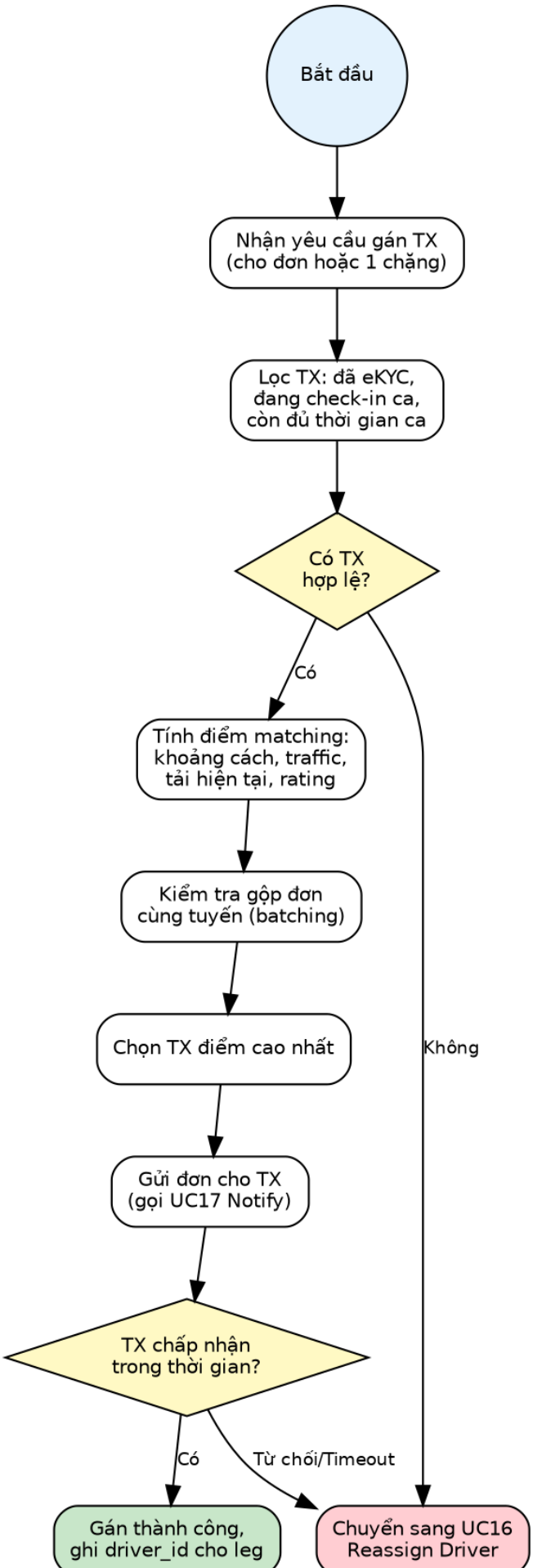
Related Screens

DSP-02	Driver Assignment
DRV-04	New Order Offer

Main Data Fields

shipment_id/leg_id, candidate_drivers[], matching_score, batch_group_id, assigned_driver_id, accept_timeout_seconds.

Activity Diagram


Bắt đầu
Nhận yêu cầu gán TX
(cho đơn hoặc 1 chặng)
Lọc TX: đã eKYC,
đang check-in ca,
còn đủ thời gian ca
Có TX
hợp lệ?
Có
Không
Tính điểm matching:
khoảng cách, traffic,
tải hiện tại, rating
Kiểm tra gộp đơn
cùng tuyến (batching)
Chọn TX điểm cao nhất
Gửi đơn cho TX
(gọi UC17 Notify)
TX chấp nhận
trong thời gian?
Có
Từ chối/Timeout
Gán thành công,
ghi driver_id cho leg
Chuyển sang UC16
Reassign Driver

Notes

The matching weight formula is defined by the team and is tunable.

UC09 — Reassign Driver

Use Case ID	UC09
Use Case Name	Reassign Driver
Module	Dispatcher / System
Objective	Automatically re-assign another driver when the current driver rejects, times out, or cancels mid-way; escalate to the dispatcher when attempts exceed the limit.
Primary Actor	System (Dispatch Engine)
Secondary Actor	Dispatcher
Preconditions	The shipment needs a driver; a reassign counter exists.
Trigger	Driver rejects / times out / cancels mid-way, or UC08 found no driver.
Postconditions	The shipment is assigned a new driver or escalated to the dispatcher; the customer's order is not cancelled.

Main Flow

51. Receive the reassign trigger (reject / timeout / mid-way cancel).
52. Record the reason and increment reassign_count.
53. If it is a mid-way cancellation, deduct driver reputation.
54. If reassign_count is within the limit, re-run UC08 (excluding the failed driver, searching around the current location).
55. If a new driver is found, assign and notify the customer of the driver change (without cancelling the order).

Alternative Flow

- A1. Mid-way cancel on a hub leg: search only within the same leg and area, not the whole journey.

Exception Flow

- E1. reassign_count exceeds the maximum → Escalate to Dispatcher.
- E2. Still no driver after retry → Escalate to Dispatcher.

Business Rules

- BR05 Driver Reassignment
- BR04 Driver Assignment

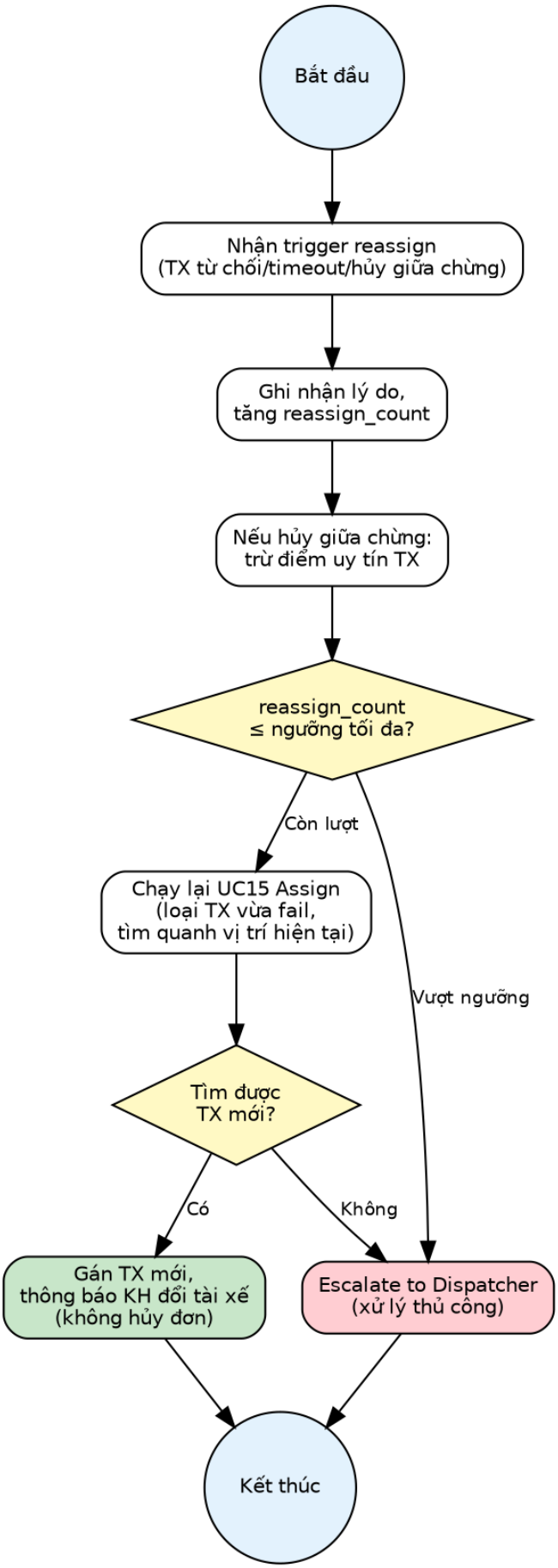
Related Screens

DSP-03	Reassignment Queue
CUS-11	Driver Change Notice
DRV-07	Cancel Order

Main Data Fields

shipment_id/leg_id, reassign_count, cancel_reason, driver_reputation_delta, reassigned_driver_id, escalated_flag.

Activity Diagram


Bắt đầu
Nhận trigger reassign
(TX từ chối/timeout/hủy giữa chừng)
Ghi nhận lý do,
tăng reassign_count
Nếu hủy giữa chừng:
trừ điểm uy tín TX
reassign_count
≤ ngưỡng tối đa?
Còn lượt
Vượt ngưỡng
Chạy lại UC15 Assign
(loại TX vừa fail,
tìm quanh vị trí hiện tại)
Tìm được
TX mới?
Có
Không
Gán TX mới,
thông báo KH đổi tài xế
(không hủy đơn)
Escalate to Dispatcher
(xử lý thủ công)
Kết thúc

Notes

Maximum reassign attempts are configurable; exceeding them must escalate instead of looping indefinitely.

UC10 — Verify Driver Identity (eKYC)

Use Case ID	UC10
Use Case Name	Verify Driver Identity (eKYC)
Module	Driver / System
Objective	Verify a driver's identity via ID card, driving licence and a liveness video, using AI and a population-database check before activating the account.
Primary Actor	Driver
Secondary Actor	System (AI)
Preconditions	The driver has a basic account (phone number).
Trigger	A new driver wants to activate their account to receive orders.
Postconditions	The driver account becomes Verified (can receive orders) or Rejected.

Main Flow

56. Driver opens the identity verification screen.
57. Driver captures the ID card (both sides) and the driving licence.
58. Driver records a liveness video.
59. System runs OCR + Face Match + Liveness (AI).
60. If the images are clear and the face matches, cross-check the population database.
61. If valid and the licence is not expired, approve and activate the driver account.

Alternative Flow

- A1. The driver retakes a photo immediately if the preview is unclear, before submitting.

Exception Flow

- E1. Blurry/low-light image → ask to retake.
- E2. Face does not match the ID → reject and log a fraud warning.
- E3. Driving licence expired → reject with reason.
- E4. Population database unreachable → keep as processing and retry.

Business Rules

- BR07 Driver eKYC
- BR16 Data Privacy

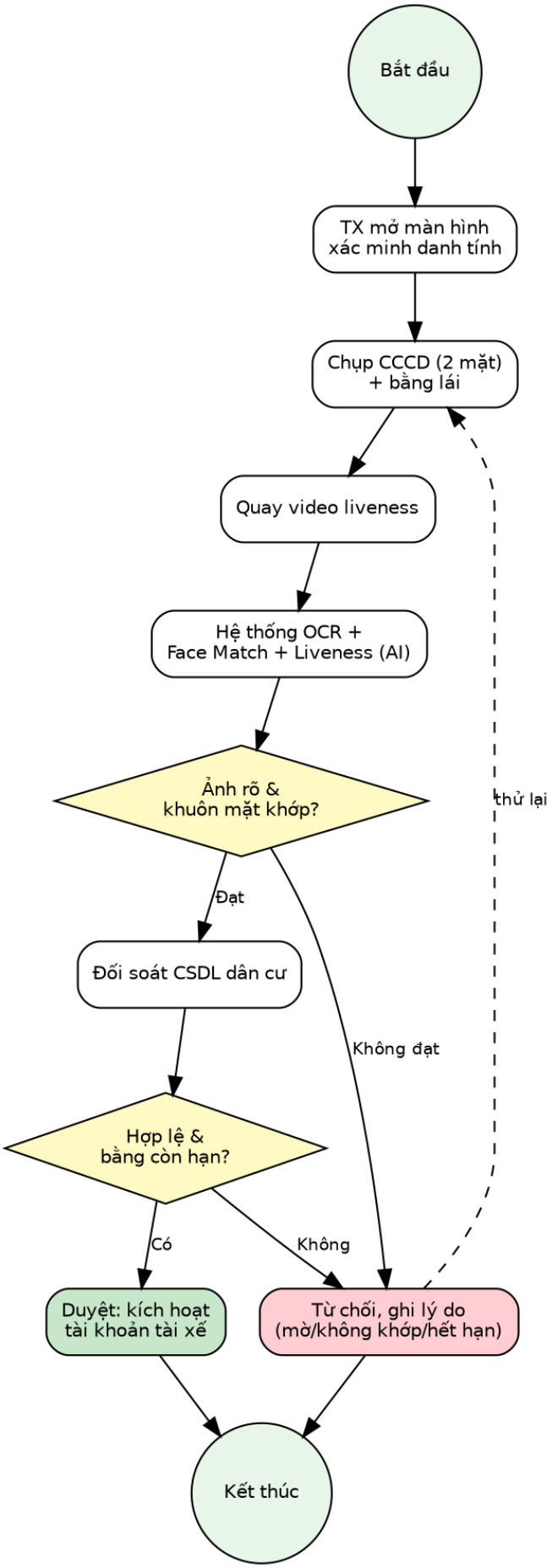
Related Screens

DRV-01	Capture ID Card
DRV-02	Capture Licence
DRV-03	Liveness Check
DRV-08	Verification Result

Main Data Fields

driver_id, id_card_front/back_image, driving_licence_image, licence_expiry_date, liveness_video, face_match_score, verification_status, rejection_reason.

Activity Diagram


Bắt đầu
TX mở màn hình xác minh danh tính
Chụp CCCD (2 mặt) + bằng lái
Quay video liveness
Hệ thống OCR + Face Match + Liveness (AI)
Ảnh rõ & khuôn mặt khớp?
thử lại
Đạt
Đối soát CSDL dân cư
Không đạt
Hợp lệ & bằng còn hạn?
Có
Không
Duyệt: kích hoạt tài khoản tài xế
Từ chối, ghi lý do (mờ/không khớp/hết hạn)
Kết thúc

Notes

A driver MUST complete eKYC before accessing operational features (UC11–UC14).

UC11 — Update Availability

Use Case ID	UC11
Use Case Name	Update Availability
Module	Driver
Objective	Let a driver declare their availability and shift (8h/12h) and starting hub, which drives the matching engine (UC08).
Primary Actor	Driver
Secondary Actor	System
Preconditions	The driver has completed eKYC (UC10).
Trigger	Start of shift, or when a driver toggles order-taking.
Postconditions	The driver's availability and shift are updated and the driver may be matched.

Main Flow

62. Driver opens the Shift screen in the app.
63. System checks that the driver is eKYC-verified.
64. Driver selects availability (online/offline).
65. Driver declares the shift (8h/12h) and the starting hub.
66. If within the allowed window, update availability so the driver becomes eligible for matching (UC08).

Alternative Flow

- A1. The driver switches shift mid-day (8h to 12h) if policy allows.

Exception Flow

- E1. Not eKYC-verified → block and require eKYC first.
- E2. Outside allowed hours / invalid shift → show error and ask to re-select.

Business Rules

- BR07 Driver eKYC
- BR11 Shift Availability

Related Screens

DRV-05	Shift & Availability

Main Data Fields

driver_id, availability_status, shift_type (8H/12H), start_hub_id, shift_start_time, shift_remaining.

Notes

Matching (UC08) only considers online drivers with enough remaining shift time for the leg.

UC12 — Accept / Reject Shipment

Use Case ID	UC12
Use Case Name	Accept / Reject Shipment
Module	Driver
Objective	Let a driver review an offered order and accept or reject it; a timeout triggers automatic reassignment.
Primary Actor	Driver
Secondary Actor	System
Preconditions	The driver is online and has been offered an order/leg.
Trigger	The driver receives a new-order notification from UC08.
Postconditions	The order is accepted (moves to pickup) or rejected/timed-out (moves to UC09).

Main Flow

67. The driver receives a new-order notification (from UC08).
68. The driver reviews the detail: pickup/drop-off, matching score, whether it is batched.
69. If the driver responds in time and accepts, the order is taken and moves to pickup.

Alternative Flow

- A1. The driver previews the route/distance before deciding.

Exception Flow

- E1. The driver rejects → trigger UC09 Reassign.
- E2. Response times out → automatically trigger UC09 Reassign.

Business Rules

- BR04 Driver Assignment
- BR05 Driver Reassignment

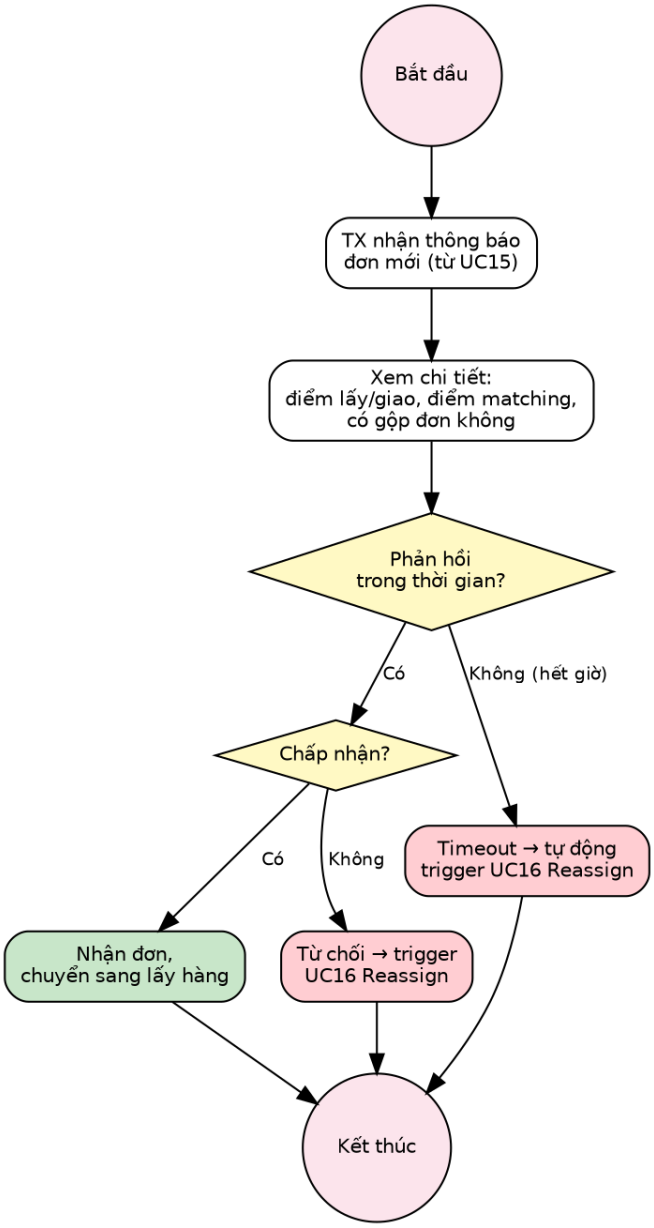
Related Screens

DRV-04	New Order Offer

Main Data Fields

driver_id, shipment_id/leg_id, matching_score, batch_flag, response (ACCEPT/REJECT/TIMEOUT), response_time.

Activity Diagram



Notes

Response time is configurable (e.g. 30s); timing out counts as a rejection.

UC13 — Confirm Pickup

Use Case ID	UC13
Use Case Name	Confirm Pickup
Module	Driver
Objective	Let a driver confirm that a parcel has been collected at each pickup point, supporting multi-pickup shipments.
Primary Actor	Driver
Secondary Actor	System
Preconditions	The driver has accepted the order and arrived at a pickup point.
Trigger	The driver arrives at a pickup point.
Postconditions	All pickup points are collected and the shipment status becomes PICKED_UP.

Main Flow

70. The driver opens the assigned order and navigates to a pickup point.
71. The driver verifies the parcel(s) at that point.
72. The driver scans/【confirms】pickup for that point.
73. If the shipment has more pickup points, the driver repeats until all are collected.
74. System updates the shipment status to PICKED_UP once all points are collected.

Alternative Flow

- A1. Partial pickup: the driver marks a point as unavailable and proceeds; the shipment is flagged for review.

Exception Flow

- E1. Parcel not ready at a pickup point → the driver records the issue and the system notifies the customer.

Business Rules

- BR03 Multiple Pickup Points
- BR10 Shipment Status

Related Screens

DRV-06	Confirm Pickup

Main Data Fields

shipment_id, pickup_points[], pickup_confirmed_flags[], picked_up_at.

Notes

For multi-pickup shipments, pickup confirmation is per point, not per shipment.

UC14 — Confirm Delivery

Use Case ID	UC14
Use Case Name	Confirm Delivery
Module	Driver
Objective	Let a driver confirm the final delivery with proof of delivery (POD) and, for COD orders, collect and reconcile the payment before closing the order.
Primary Actor	Driver
Secondary Actor	System
Preconditions	The shipment is out for delivery (OUT_FOR_DELIVERY).
Trigger	The driver arrives at the final delivery point.
Postconditions	The shipment becomes COMPLETED; any COD transaction is recorded; the customer is invited to rate.

Main Flow

75. The driver arrives at the final delivery point.
76. The driver taps "Delivered" and captures a POD photo.
77. If the order is COD, the driver collects the cash and enters the amount.
78. If the amount matches, the system sets the status to COMPLETED.
79. System notifies the customer (UC19) and invites a driver rating.

Alternative Flow

- A1. Recipient absent: the system allows completion with proof after a waiting period, per policy.

Exception Flow

- E1. COD amount mismatch → prompt to re-enter/verify.
- E2. POD photo cannot be captured → require a retake before completing.

Business Rules

- BR12 Proof of Delivery
- BR13 COD Reconciliation
- BR14 Notification

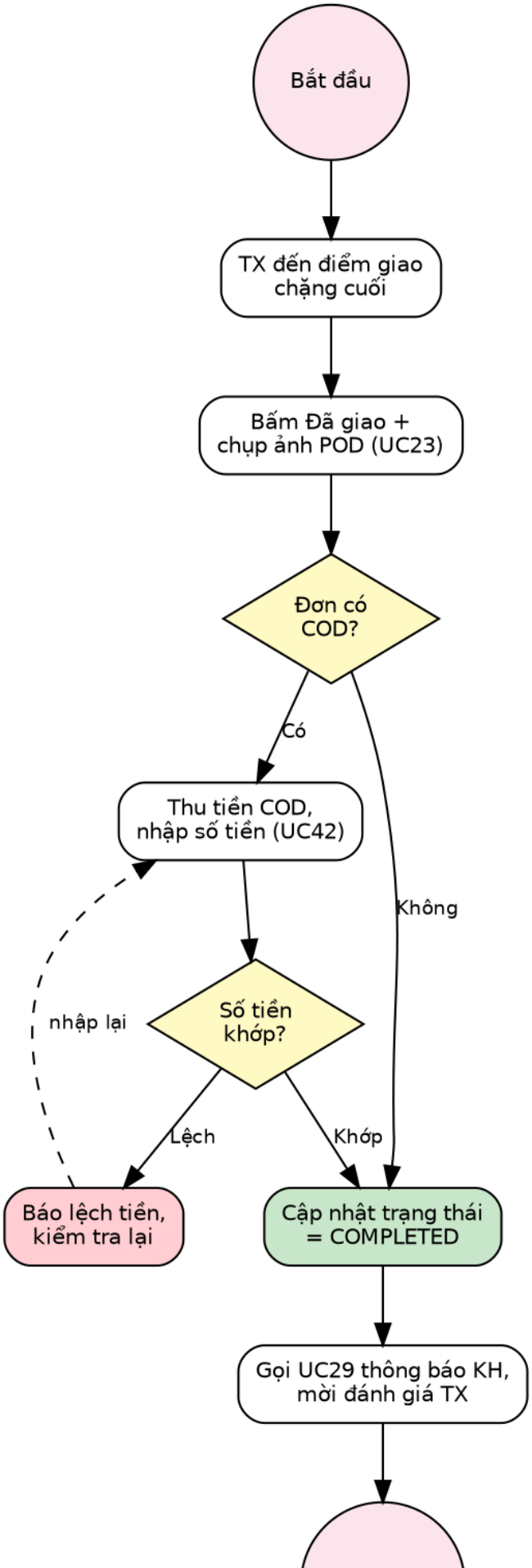
Related Screens

DRV-09	Confirm Delivery
DRV-10	Collect COD
DRV-11	Delivery Completed

Main Data Fields

shipment_id, proof_of_delivery_image, payment_method, cod_amount, amount_collected, payment_status, completed_at, customer_rating.

Activity Diagram


Bắt đầu
TX đến điểm giao chặng cuối
Bấm Đã giao + chụp ảnh POD (UC23)
Đơn có COD?
Có
Không
Thu tiền COD, nhập số tiền (UC42)
Số tiền khớp?
nhập lại
Lệch
Khớp
Báo lệch tiền, kiểm tra lại
Cập nhật trạng thái = COMPLETED
Gọi UC29 thông báo KH, mời đánh giá TX

Notes

A COD order can only be closed when the collected amount matches the amount due.

UC15 — Receive Shipment at Hub

Use Case ID	UC15
Use Case Name	Receive Shipment at Hub
Module	Hub Staff / Driver
Objective	Record parcels arriving at a hub via scanning, verify their condition, and set the status to AT_HUB.
Primary Actor	Hub Staff
Secondary Actor	Driver / System
Preconditions	The parcel is in transit to the hub; staff/driver has a scanning device.
Trigger	A first-mile driver brings parcels to the hub.
Postconditions	The parcel status becomes AT_HUB and the customer is notified.

Main Flow

80. Hub Staff/driver scans the parcel code at the hub.
81. System checks that the code is valid and for the correct hub.
82. Staff verifies the parcel information against the system.
83. If the parcel is intact, update the status to AT_HUB.
84. Call UC19 to notify the customer that the parcel has arrived at the hub.

Alternative Flow

- A1. Batch scan: staff scan multiple parcels in one receiving session.

Exception Flow

- E1. Invalid code or wrong hub → show error and do not receive.
- E2. Damaged/incomplete parcel → record the incident and move to exception handling.

Business Rules

- BR06 Hub Routing
- BR10 Shipment Status
- BR18 Hub Traceability

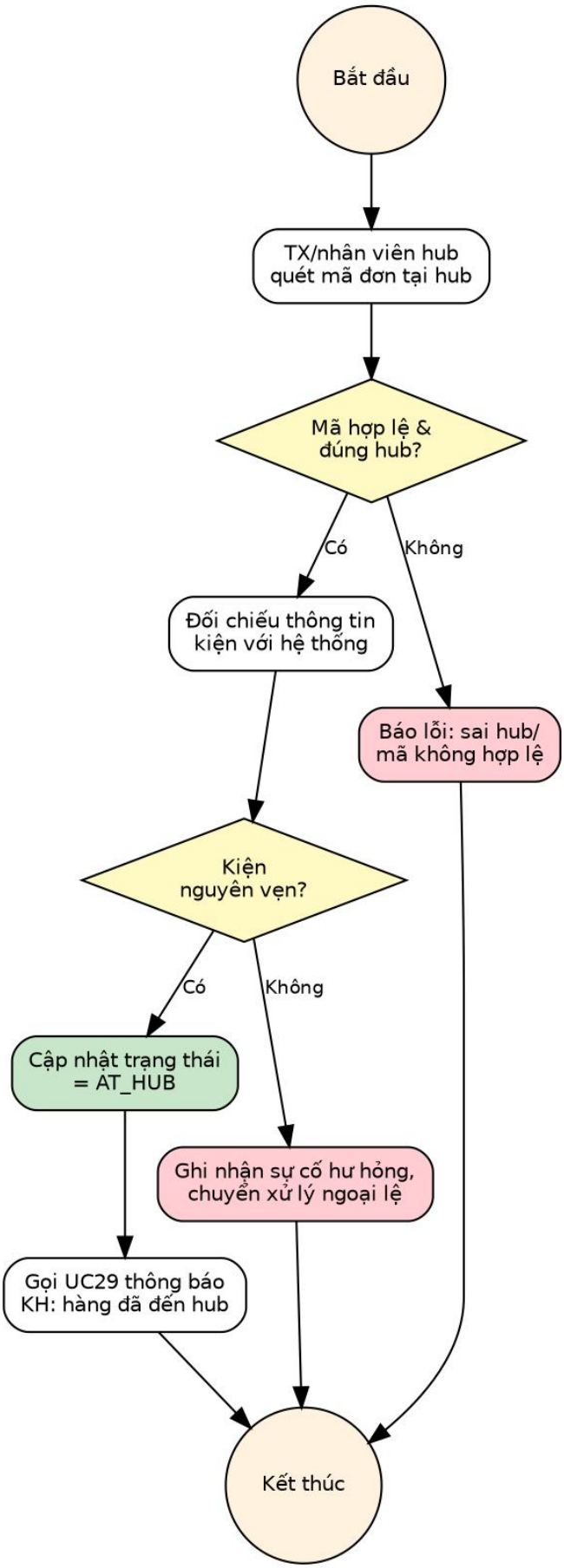
Related Screens

HUB-01	Receive at Hub
HUB-04	Parcel Incident

Main Data Fields

shipment_id, leg_id, hub_id, scan_time, parcel_condition, status (AT_HUB).

Activity Diagram


Bắt đầu
TX/nhân viên hub quét mã đơn tại hub
Mã hợp lệ & đúng hub?
Có
Không
Đối chiếu thông tin kiện với hệ thống
Báo lỗi: sai hub/ mã không hợp lệ
Kiện nguyên vẹn?
Có
Không
Cập nhật trạng thái = AT_HUB
Ghi nhận sự cố hư hỏng, chuyển xử lý ngoại lệ
Gọi UC29 thông báo KH: hàng đã đến hub
Kết thúc

Notes

Every custody/status change at a hub must be scanned for traceability.

UC16 — Sort & Consolidate Shipment

Use Case ID	UC16
Use Case Name	Sort & Consolidate Shipment
Module	Hub Staff / System
Objective	Sort parcels at a hub by destination hub/route and consolidate same-route parcels into batches to optimise trips.
Primary Actor	Hub Staff
Secondary Actor	System
Preconditions	Parcels are in the AT_HUB status.
Trigger	There are AT_HUB parcels to process.
Postconditions	Same-route parcels are grouped into a batch_group ready to leave the hub.

Main Flow

85. Retrieve the list of AT_HUB parcels.
86. Sort them by destination hub/route.
87. Consolidate same-route parcels into a group (batch).
88. When the batch reaches sufficient volume or the cut-off time, create a batch_group ready for dispatch (UC17).

Alternative Flow

- A1. Priority parcels are placed in the earliest outgoing batch instead of waiting for a full batch.

Exception Flow

- E1. Not enough volume and cut-off not reached → hold parcels and keep consolidating.

Business Rules

- BR06 Hub Routing
- BR02 Priority Delivery
- BR18 Hub Traceability

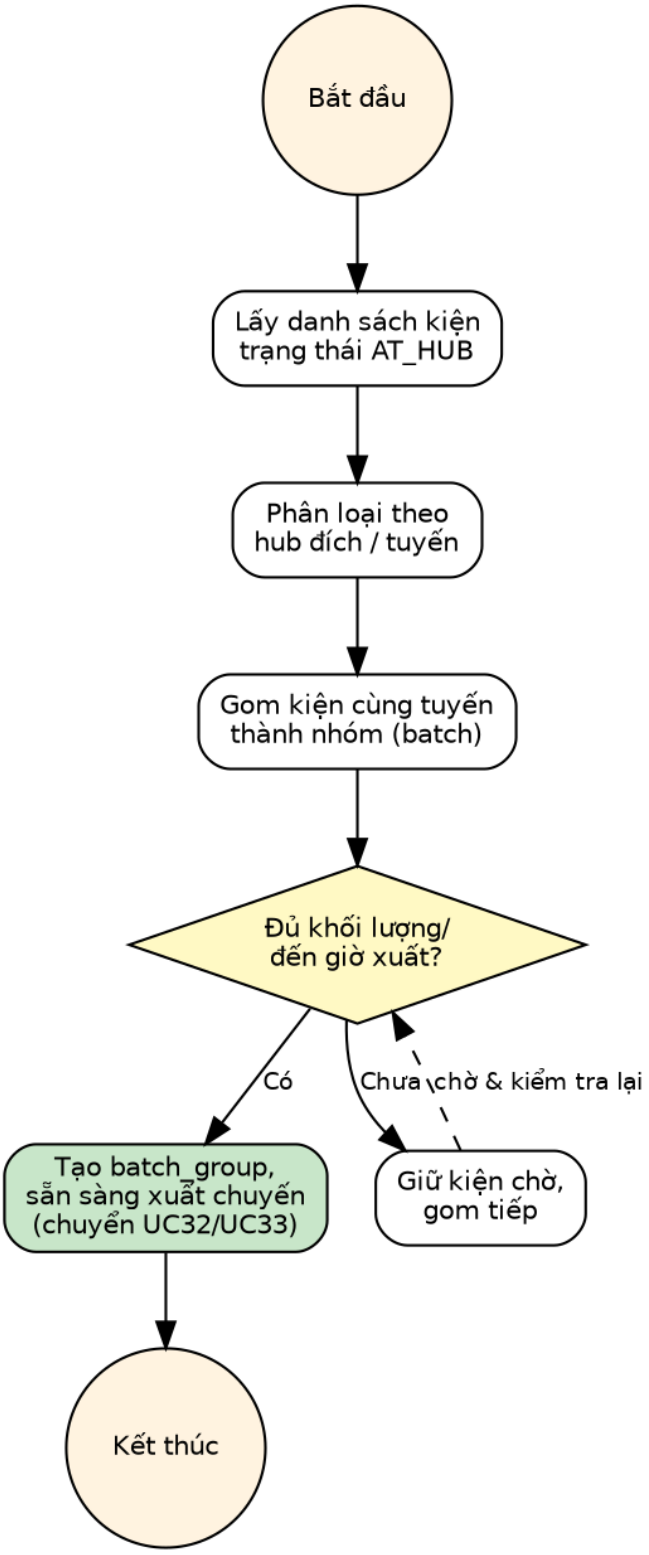
Related Screens

HUB-02	Sort & Consolidate
HUB-05	Batch Management

Main Data Fields

shipment_id, destination_hub_id, route_id, batch_group_id, batch_status, cutoff_time.

Activity Diagram



Notes

Batch dispatch conditions (minimum volume, cut-off) are configurable per hub.

UC17 — Dispatch from Hub

Use Case ID	UC17
Use Case Name	Dispatch from Hub
Module	Hub Staff / System
Objective	Hand a batch/parcels from the hub to the next leg: assign a last-mile driver (via UC08) or create a line-haul trip, with a scan-verified handoff.
Primary Actor	Hub Staff
Secondary Actor	System / Driver
Preconditions	A batch is consolidated and ready to leave.
Trigger	A batch is ready to leave the hub (after UC16).
Postconditions	Parcels move to OUT_FOR_DELIVERY (last-mile) or IN_TRANSIT (line-haul) with a handoff record.

Main Flow

89. Receive a batch that is ready to leave the hub.
90. Determine whether the next leg is last-mile or line-haul.
91. If last-mile, call UC08 to assign a driver checked-in at this hub.
92. If line-haul, create a hub-to-hub trip.
93. Scan the handoff to the driver/trip.
94. If the handoff matches the code and count, update the status to OUT_FOR_DELIVERY / IN_TRANSIT.

Alternative Flow

- A1. Partial handoff: if one driver cannot carry the whole batch, split the remainder to another driver/trip.

Exception Flow

- E1. Parcel-count mismatch at handoff → recount before confirming.

Business Rules

- BR06 Hub Routing
- BR18 Hub Traceability
- BR04 Driver Assignment

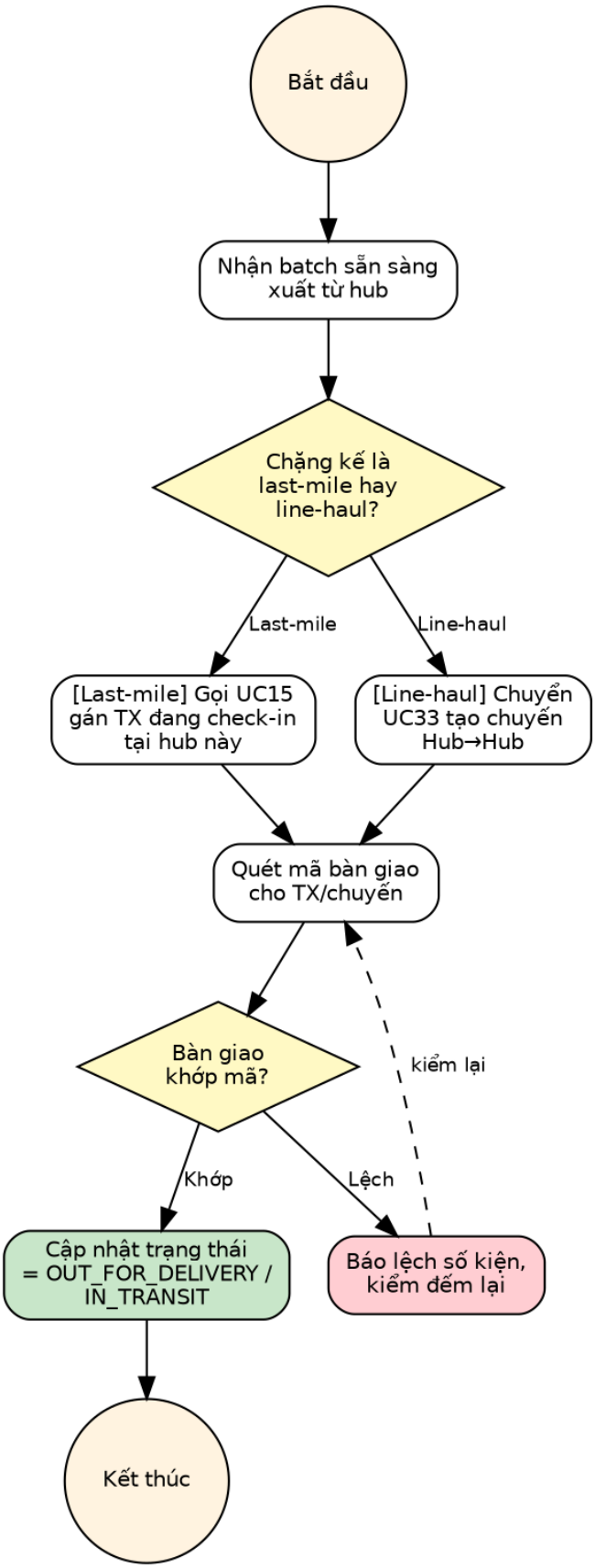
Related Screens

HUB-03	Dispatch / Handoff
DRV-12	Receive from Hub

Main Data Fields

batch_group_id, next_leg_type (LAST_MILE/LINE_HAUL), assigned_driver_id/trip_id, scanned_count, status.

Activity Diagram


Bắt đầu
Nhận batch sẵn sàng xuất từ hub
Chặng kế là last-mile hay line-haul?
Last-mile
Line-haul
[Last-mile] Gọi UC15 gán TX đang check-in tại hub này
[Line-haul] Chuyển UC33 tạo chuyến Hub→Hub
Quét mã bàn giao cho TX/chuyến
Bàn giao khớp mã?
kiểm lại
Khớp
Lệch
Cập nhật trạng thái = OUT_FOR_DELIVERY / IN_TRANSIT
Báo lệch số kiện, kiểm đếm lại
Kết thúc

Notes

A last-mile driver must be checked-in at the correct destination hub.

UC18 — Recommend Optimal Hub (AI)

Use Case ID	UC18
Use Case Name	Recommend Optimal Hub (AI)
Module	System
Objective	When a shipment is routed through a hub, recommend the optimal transit hub(s) using AI based on location, hub load, distance and SLA.
Primary Actor	System (AI module)
Secondary Actor	Dispatcher
Preconditions	There is a list of active hubs with load and coordinate data.
Trigger	UC07 sets the strategy to HUB.
Postconditions	A recommended hub is returned; UC07 uses it to route the shipment.

Main Flow

95. Receive the hub-recommendation request from UC07.
96. Collect data: order location, current load per hub, distances, service SLA.
97. The AI model scores each candidate hub.
98. If a hub meets the confidence threshold, select the highest-scoring hub.
99. Return the recommendation (first-hub, last-hub) with a confidence score to UC07.

Alternative Flow

- A1. Priority order: the AI prefers a hub with shorter transit time even at slightly higher load.

Exception Flow

- E1. No hub meets the threshold → fall back to the nearest hub by distance and flag for the dispatcher.
- E2. Hub load data is stale → use the latest available data and log a warning.

Business Rules

- BR06 Hub Routing

Related Screens

DSP-04	Hub Recommendation

Main Data Fields

shipment_id, candidate_hubs[], hub_load, hub_distance, sla, recommended_first_hub, recommended_last_hub, confidence_score.

Notes

AI only advises; the dispatcher can override. A non-AI fallback must always exist so orders never stall.

UC19 — Send Notification

Use Case ID	UC19
Use Case Name	Send Notification
Module	System
Objective	Send status and event notifications to customers and drivers through the appropriate channel.
Primary Actor	System
Secondary Actor	Customer / Driver
Preconditions	A notifiable event has occurred (status change, driver change, delivery, etc.).
Trigger	A business event requiring notification is raised by another use case.
Postconditions	The recipient is notified and the result is logged.

Main Flow

100. Receive a notification event with its type and recipient.
101. Select the template and channel (push/SMS/email) based on the event type.
102. Render the message from the template and data.
103. Send the message and record the delivery result.

Alternative Flow

- A1. If the primary channel fails, fall back to a secondary channel.

Exception Flow

- E1. All channels fail → log the failure for later retry.

Business Rules

- BR14 Notification
- BR17 Audit Logging

Related Screens

SYS-02	Notification Log (Admin)

Main Data Fields

event_type, recipient_id, channel, template_id, message_body, sent_status, sent_at.

Notes

This is an included use case invoked by UC14, UC15 and others; it has no direct end-user screen.

UC20 — Generate Tracking Number

Use Case ID	UC20
Use Case Name	Generate Tracking Number
Module	System
Objective	Generate a unique tracking number for a shipment when it is created.
Primary Actor	System
Secondary Actor	Customer / Counter Staff
Preconditions	A shipment has passed validation.
Trigger	A shipment is created (UC03/UC06).
Postconditions	The shipment has a unique tracking number.

Main Flow

104. Receive the validated shipment.
105. Generate a unique tracking number following the numbering rule.
106. Ensure the number is unique across the system.
107. Attach the tracking number to the shipment and return it.

Alternative Flow

None.

Exception Flow

- E1. Rare collision detected → regenerate until unique.

Business Rules

- BR09 Tracking Number
- BR10 Shipment Status

Related Screens

(none)	Runs in the background; the number appears on UC03/UC06 confirmation screens.

Main Data Fields

shipment_id, tracking_number, generated_at.

Notes

Included by UC03 and UC06; format follows BR09.

Chapter 8. Functional Requirements

ID	Functional Requirement
FR-001	The system shall allow a customer to register an account with OTP verification.
FR-002	The system shall allow all human actors to log in according to their role.
FR-003	The system shall allow a customer to reset a forgotten password.
FR-004	The system shall allow a customer to create a delivery request with one or more pickup points.
FR-005	The system shall allow a customer to specify a single destination per shipment.
FR-006	The system shall allow a customer to choose Standard or Priority service.
FR-007	The system shall allow a customer to mark an order as COD or prepaid.
FR-008	The system shall validate shipment input before creation.
FR-009	The system shall generate a unique tracking number for every shipment.
FR-010	The system shall allow a customer to track a shipment in real time.
FR-011	The system shall allow a customer to view their shipment history.
FR-012	The system shall allow counter staff to create a walk-in shipment on behalf of a customer.
FR-013	The system shall allow counter staff to look up or quickly register a walk-in customer.
FR-014	The system shall automatically determine whether a shipment is Direct or Hub.
FR-015	The system shall allow a dispatcher to configure routing thresholds.
FR-016	The system shall recommend an optimal hub using AI for hub-routed shipments.
FR-017	The system shall provide a non-AI fallback for hub selection.
FR-018	The system shall match a shipment to the best-scoring eligible driver.
FR-019	The system shall consider distance, traffic, current load and rating in driver matching.
FR-020	The system shall batch same-route orders where eligible.
FR-021	The system shall allow a dispatcher to override an automated assignment.
FR-022	The system shall reassign a driver on reject, timeout or mid-way cancellation.
FR-023	The system shall cap reassignment attempts and escalate to the dispatcher when exceeded.
FR-024	The system shall reduce driver reputation for mid-way cancellations.
FR-025	The system shall require driver eKYC before granting operational access.
FR-026	The system shall run OCR, face match and liveness detection during eKYC.
FR-027	The system shall cross-check driver identity against the population database.
FR-028	The system shall allow a driver to set availability and declare a shift and starting hub.
FR-029	The system shall only match drivers with enough remaining shift time.
FR-030	The system shall allow a driver to accept or reject an offered order within a time window.
FR-031	The system shall treat a response timeout as a rejection.
FR-032	The system shall allow a driver to confirm pickup per pickup point.
FR-033	The system shall allow a driver to capture proof of delivery at completion.
FR-034	The system shall allow a driver to collect and reconcile COD before completing an order.
FR-035	The system shall allow hub staff to receive parcels at a hub by scanning.
FR-036	The system shall allow hub staff to sort and consolidate parcels into batches.
FR-037	The system shall allow hub staff to dispatch a batch to the next leg with a scanned handoff.
FR-038	The system shall update and expose shipment status throughout the lifecycle.
FR-039	The system shall notify customers and drivers on key events.
FR-040	The system shall log key business actions for audit.

Chapter 9. Non-Functional Requirements

9.1 Performance

- Driver matching for a shipment/leg shall complete within 2 seconds.
- eKYC verification results shall return within 30 seconds under normal load.
- Tracking status shall refresh within 5 seconds of a status change.

9.2 Availability

- The dispatch and matching services shall target 99.5% uptime.
- On reassignment, a replacement driver shall be found within 15 seconds where one exists.

9.3 Security

- Identity and personal data shall be encrypted in transit (TLS 1.2+) and at rest.
- Access shall be role-based; each actor sees only permitted features.
- Authentication shall lock an account after a configured number of failed attempts.

9.4 Scalability

- The matching service shall handle at least 1,000 concurrent requests at peak.
- Hub operations shall scale to multiple hubs without redesign.

9.5 Audit Logging

- Assignment, reassignment, hub handoff, delivery and notification events shall be logged.
- Logs shall be retained for a configurable period for review.

Chapter 10. Screen Mapping

Each use case maps to the screens a designer must produce. Screen IDs follow the inventory in Chapter 11.

Use Case	Screen(s)
UC01	CUS-08, CUS-09
UC02	CUS-10, SYS-01
UC03	CUS-01, CUS-02, CUS-03, CUS-04, CUS-05
UC04	CUS-06
UC05	CUS-07
UC06	CS-01, CS-02
UC07	DSP-01, DSP-05
UC08	DSP-02, DRV-04
UC09	DSP-03, CUS-11, DRV-07
UC10	DRV-01, DRV-02, DRV-03, DRV-08
UC11	DRV-05
UC12	DRV-04
UC13	DRV-06
UC14	DRV-09, DRV-10, DRV-11
UC15	HUB-01, HUB-04
UC16	HUB-02, HUB-05
UC17	HUB-03, DRV-12
UC18	DSP-04
UC19	SYS-02
UC20	(background)

Chapter 11. Screen Inventory

11.1 Customer App

Screen ID	Screen Name	Description	Primary User
CUS-01	Home	Entry screen with quick actions (create, track, history).	Customer
CUS-02	Create Delivery	Start a new delivery request.	Customer
CUS-03	Pickup Information	Enter one or more pickup points.	Customer
CUS-04	Receiver Information	Enter the single destination and parcel/COD info.	Customer
CUS-05	Confirmation	Review, choose service, pay if prepaid, show tracking number.	Customer
CUS-06	Track Shipment	Show real-time status and location.	Customer
CUS-07	Shipment History	List past and completed shipments.	Customer
CUS-08	Register	Create a new account.	Customer
CUS-09	OTP Verification	Enter the OTP to verify registration.	Customer
CUS-10	Login	Authenticate into the app.	Customer
CUS-11	Driver Change Notice	Inform the customer their driver changed.	Customer

11.2 Counter Portal

Screen ID	Screen Name	Description	Primary User
CS-01	Walk-in Shipment	Create a shipment for a walk-in customer.	Counter Staff
CS-02	Customer Lookup	Find or quickly register a walk-in customer.	Counter Staff

11.3 Dispatcher Portal

Screen ID	Screen Name	Description	Primary User
DSP-01	Dispatch Board	Overview of shipments and their strategy/status.	Dispatcher
DSP-02	Driver Assignment	Monitor and override driver assignments.	Dispatcher
DSP-03	Reassignment Queue	Handle escalated/failed assignments.	Dispatcher
DSP-04	Hub Recommendation	View AI hub suggestions and override.	Dispatcher
DSP-05	Routing Configuration	Configure distance/shift thresholds.	Dispatcher

11.4 Driver App

Screen ID	Screen Name	Description	Primary User
DRV-01	Capture ID Card	Photograph the ID card for eKYC.	Driver
DRV-02	Capture Licence	Photograph the driving licence.	Driver
DRV-03	Liveness Check	Record the liveness video.	Driver
DRV-04	New Order Offer	Review and accept/reject an offered order.	Driver
DRV-05	Shift & Availability	Toggle availability and declare shift/hub.	Driver
DRV-06	Confirm Pickup	Confirm pickup per pickup point.	Driver
DRV-07	Cancel Order	Cancel with a reason (reputation warning).	Driver
DRV-08	Verification Result	Show eKYC success/failure and reason.	Driver
DRV-09	Confirm Delivery	Mark delivered and capture POD.	Driver
DRV-10	Collect COD	Enter and confirm collected COD amount.	Driver
DRV-11	Delivery Completed	Completion summary.	Driver
DRV-12	Receive from Hub	Confirm parcels received from a hub.	Driver

11.5 Hub Portal

Screen ID	Screen Name	Description	Primary User
HUB-01	Receive at Hub	Scan and receive incoming parcels.	Hub Staff
HUB-02	Sort & Consolidate	Sort parcels by route and build batches.	Hub Staff
HUB-03	Dispatch / Handoff	Scan handoff to driver or line-haul trip.	Hub Staff
HUB-04	Parcel Incident	Record damaged/incorrect parcels.	Hub Staff
HUB-05	Batch Management	View and release consolidated batches.	Hub Staff

11.6 System / Shared

Screen ID	Screen Name	Description	Primary User
SYS-01	Forgot Password	Reset password flow.	All
SYS-02	Notification Log	Admin view of sent notifications.	Operations